Data Description

This data set is collected through google form from 102 peoples. The dataset consists of survey responses collected to analyse the relationship between sleep habits and mood. It includes variables related to sleep duration, bedtime patterns, screen exposure before sleep, sleep quality, and emotional well-being.

Key variables in the dataset include:

1. **Response_No -** Response number (Numerical)
2. **Name -** Name of Responder
3. **Age_Group -** Age group (Categorical)
4. **Sleep_Duration -** Hours of sleep do you usually get on a typical night. (Categorical)
5. **Bedtime_Range -** Time do you usually go to bed. (Categorical)
6. **Difficulty_falling_asleep** - Difficulty falling asleep. (Categorical)
7. **Mood_Rating -** Rate your mood today. (Categorical)
8. **Sleep_Aids_Usage -** Use of any sleep aids. (Categorical)
9. **Daily_Nap_Count -** Number of Naps do you take during the day. (Categorical)
10. **Morning_Refresh_Score** - How Refreshed do you feel when you wake up. (Categorical)
11. **Sleep_Screen_Time_Category -** On average, how many hours do you spend using electronic devices before sleeping. (Categorical)

The dataset contains both numerical and categorical variables, enabling univariate, bivariate, and multivariate analysis to study the impact of sleep patterns on mood and sleep hygiene.

Data Analysis

Q1. Crated a Exploratory Data Analysis (EDA) table. (DESCRIPTIVE STATISTICS)

In [13]:
```
# Calculate values
avg_mood = sleep_and_mood_df['Mood_Rating'].mean()
avg_refresh = sleep_and_mood_df['Morning_Refresh_Score'].mean()
common_sleep_duration = sleep_and_mood_df['Sleep_Duration'].mode()[0]
common_bedtime = sleep_and_mood_df['Bedtime_Range'].mode()[0]
difficulty_percent = (sleep_and_mood_df['Difficulty_falling_asleep'] == 'Yes').mean() * 100
avg_naps = sleep_and_mood_df['Daily_Nap_Count'].mean()
common_device_usage = sleep_and_mood_df['Pre_Sleep_Screen_Time_Category'].mode()[0]
sleep_aid_counts = sleep_and_mood_df['Sleep_Aids_Usage'].value_counts().to_dict()
```

```
# Create summary table
summary_table = pd.DataFrame({
    "Metric": [
        "Average Mood Rating",
        "Average Refreshment Score",
        "Most Common Sleep Duration",
        "Most Common Bedtime Range",
        "% With Difficulty Falling Asleep",
        "Average Naps Per Day",
        "Most Common Device Usage Before Bed",
        "Sleep Aid Usage Count"
    ],
    "Value": [
        round(avg_mood, 2),
        round(avg_refresh, 2),
        common_sleep_duration,
        common_bedtime,
        round(difficulty_percent, 2),
        round(avg_naps, 2),
        common_device_usage,
        sleep_aid_counts
    ]
})

summary_table
```

Output:

	Metric	Value
0	Average Mood Rating	4.07
1	Average Refreshment Score	3.73
2	Most Common Sleep Duration	6 to 8 hours
3	Most Common Bedtime Range	11:00 PM – 12:59 AM
4	% With Difficulty Falling Asleep	25.49
5	Average Naps Per Day	0.43
6	Most Common Device Usage Before Bed	more than 2 hours
7	Sleep Aid Usage Count	{'no': 99, 'Yes': 3}

Analytical statement: As part of Exploratory Data Analysis (EDA), descriptive statistics were calculated to understand overall sleep behaviour patterns. The average mood rating (4.07) and refreshment score (3.73) indicate generally positive well-being among respondents. The most common sleep duration is 6–8 hours, suggesting that most participants maintain recommended sleep levels.

However, 25.49% of respondents report difficulty falling asleep, and the most common device usage before bed exceeds two hours, indicating potential behavioural factors affecting sleep quality. Despite this, sleep aid usage is minimal, with only 3 respondents reporting usage.

These findings provide foundational insights into sleep habits and set the stage for deeper correlation and multivariate analysis.

Q2. Group respondents by age group and calculate average sleep duration.

```
In [22]: sleep_and_mood_df.groupby('Age_Group')['Sleep_Duration'].value_counts()
```

```
Out[22]: Age_Group     Sleep_Duration
         15-18 years   6 to 8 hours          8
                       4 to 6 hours          4
                       Less than 4 hours     1
                       More than 8 hours     1
         19- 25 years  6 to 8 hours         29
                       More than 8 hours    17
                       4 to 6 hours         10
         26- 35 years  More than 8 hours    12
                       6 to 8 hours         10
         35+ years     More than 8 hours     8
                       6 to 8 hours          2
         Name: count, dtype: int64
```

Analytical statement: The analysis shows that younger respondents (15–25 years) most commonly sleep 6 to 8 hours, while older age groups (26+ years) tend to sleep more than 8 hours.

This indicates a shift toward longer sleep duration as age increases, suggesting that sleep patterns vary across different age groups.

Q3. Group by bedtime category and calculate average mood rating.

```
In [23]: sleep_and_mood_df.groupby('Bedtime_Range')['Mood_Rating'].mean()
```

```
Out[23]: Bedtime_Range
         11:00 PM - 12:59 AM    3.962264
         9:00 PM - 10:59 PM     4.411765
         After 1:00 AM          3.454545
         Before 9:00 PM         4.250000
         Name: Mood_Rating, dtype: float64
```

Analytical statement: The analysis of average mood rating across bedtime categories reveals a clear pattern linking earlier sleep times with better mood outcomes.

Respondents who went to bed between 9:00 PM – 10:59 PM reported the highest average mood rating (4.41), followed closely by those sleeping before 9:00 PM (4.25). In contrast, individuals who slept between 11:00 PM – 12:59 AM had a lower

average mood rating (3.96), and those sleeping after 1:00 AM reported the lowest mood score (3.45).

This indicates a negative relationship between late bedtime and mood level. As bedtime becomes later, average mood rating decreases. The findings suggest that maintaining an earlier sleep schedule is associated with better emotional well-being.

Q4. Calculate average refreshment score based on sleep duration.

```
In [24]: sleep_and_mood_df.groupby('Sleep_Duration')['Morning_Refresh_Score'].mean()

Out[24]: Sleep_Duration
         4 to 6 hours         2.785714
         6 to 8 hours         3.714286
         Less than 4 hours    2.000000
         More than 8 hours    4.131579
         Name: Morning_Refresh_Score, dtype: float64
```

Analytical statement: The analysis shows a strong positive relationship between sleep duration and morning refreshment levels.

Respondents who slept more than 8 hours reported the highest average refresh score (4.13), followed by those sleeping 6 to 8 hours (3.71). In contrast, individuals who slept 4 to 6 hours had a lower refresh score (2.79), while those with less than 4 hours of sleep reported the lowest refreshment level (2.00).

This indicates that as sleep duration increases, the feeling of refreshment after waking up also improves. The findings suggest that insufficient sleep significantly reduces morning energy levels, while adequate sleep enhances overall restfulness.

Q5. Group by device usage before sleep and calculate average sleep quality.

```
In [25]: sleep_and_mood_df.groupby('Pre_Sleep_Screen_Time_Category')['Morning_Refresh_Score'].mean()

Out[25]: Pre_Sleep_Screen_Time_Category
         1 to 2 hours         3.863636
         less than 1 hour     3.400000
         more than 2 hours    3.689189
         none                 5.000000
         Name: Morning_Refresh_Score, dtype: float64
```

Analytical statement: The analysis shows that respondents with no screen exposure before bed reported the highest morning refresh score (5.00). Among those who used devices, individuals with 1 to 2 hours of screen time had a slightly higher refresh score (3.86) compared to those with more than 2 hours (3.69) and less than 1 hour (3.40).

Overall, the findings suggest that avoiding screen usage before bedtime is associated with better morning refreshment, while increased screen exposure may slightly reduce how refreshed individuals feel after waking up.

Q6. Count number of respondents by difficulty falling asleep (Yes/No).

```
In [26]: sleep_and_mood_df['Difficulty_falling_asleep'].value_counts()
```

```
Out[26]: Difficulty_falling_asleep
         No     76
         Yes    26
         Name: count, dtype: int64
```

Analytical statement: Out of the total respondents, 76 individuals reported not having difficulty falling asleep, while 26 respondents reported experiencing difficulty.

This indicates that although the majority of participants do not face sleep initiation issues, a significant portion (about one-fourth of the sample) experiences difficulty falling asleep, highlighting a notable concern in sleep patterns within the dataset.

Q7. Group by sleep aid usage and calculate average mood.

```
In [27]: sleep_and_mood_df.groupby('Sleep_Aids_Usage')['Mood_Rating'].mean()
```

```
Out[27]: Sleep_Aids_Usage
         Yes    4.333333
         no     4.060606
         Name: Mood_Rating, dtype: float64
```

Analytical statement: The analysis shows that respondents who use sleep aids reported a slightly higher average mood rating (4.33) compared to those who do not use sleep aids (4.06).

Although the difference is small, this may suggest that individuals using sleep aids experience marginally better mood levels, possibly due to improved sleep quality. However, further analysis would be needed to determine whether this relationship is statistically significant.

Q8. Create a summary table showing age group vs average refreshment score.

```
In [28]: sleep_and_mood_df.groupby('Age_Group')['Morning_Refresh_Score'].mean().reset_index()
```

Out[28]:

	Age_Group	Morning_Refresh_Score
0	15-18 years	2.857143
1	19- 25 years	3.660714
2	26- 35 years	4.136364
3	35+ years	4.400000

Analytical statement: The analysis shows a clear increasing trend in refreshment scores with age. Respondents aged 15–18 years reported the lowest average refreshment score (2.86), followed by the 19–25 years group (3.66). In contrast, participants aged 26–35 years (4.14) and 35+ years (4.40) reported significantly higher refreshment levels.

This suggests that older respondents tend to feel more refreshed after waking up compared to younger individuals. The results indicate a possible improvement in sleep quality or sleep habits with increasing age.

Q9. Determine which bedtime category is associated with the lowest mood rating.

```
In [29]: sleep_and_mood_df.groupby('Bedtime_Range')['Mood_Rating'].mean().sort_values()

Out[29]: Bedtime_Range
         After 1:00 AM          3.454545
         11:00 PM - 12:59 AM    3.962264
         Before 9:00 PM         4.250000
         9:00 PM - 10:59 PM     4.411765
         Name: Mood_Rating, dtype: float64
```

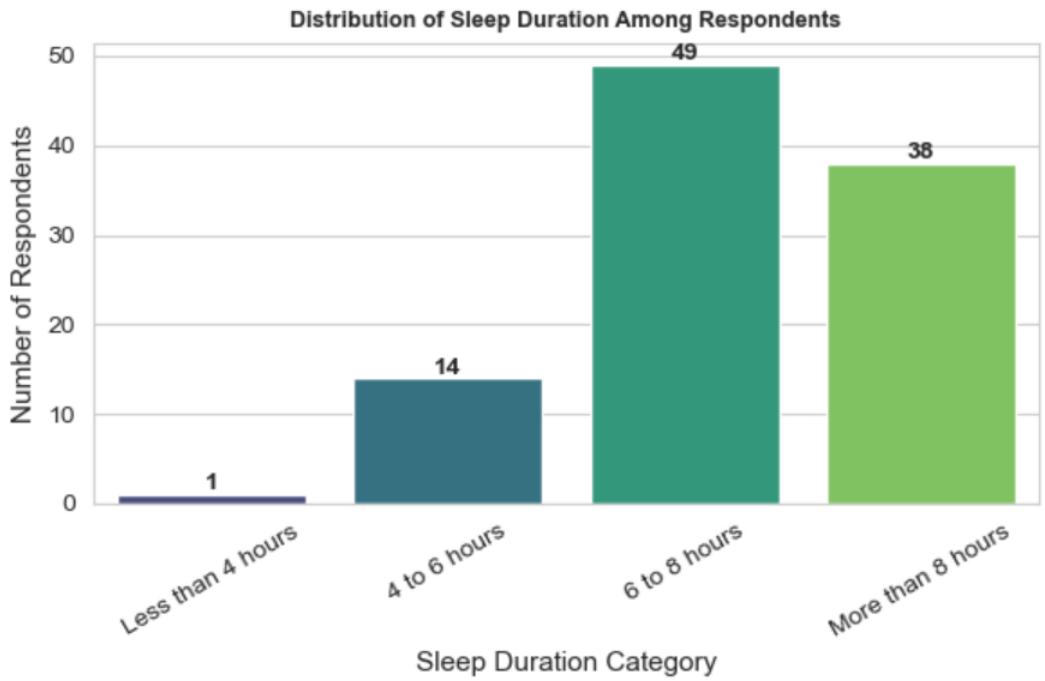
Analytical statement: The analysis indicates that respondents who go to bed after 1:00 AM have the lowest average mood rating (3.45). Mood ratings gradually improve as bedtime becomes earlier. Participants who sleep before 9:00 PM (4.25) and between 9:00 PM – 10:59 PM (4.41) report the highest mood scores.

This suggests a strong relationship between earlier bedtime and better mood outcomes. Late-night sleeping habits appear to be associated with lower emotional well-being.

Data Visualization

Q1. What is the distribution of Sleep Duration among respondents?

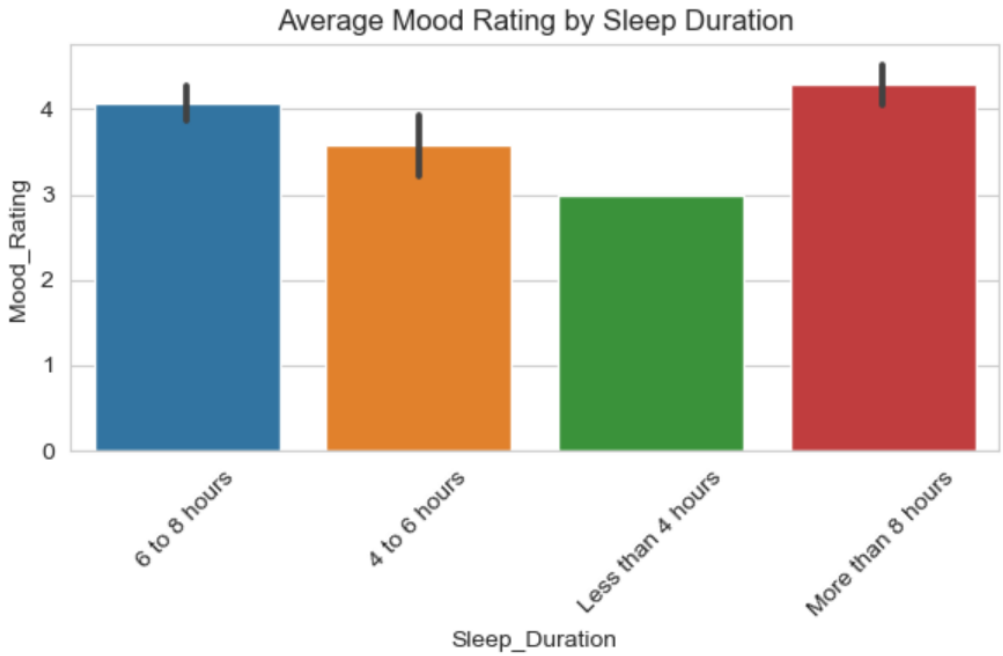
```
In [38]: # Define desired order
order = ['Less than 4 hours','4 to 6 hours','6 to 8 hours','More than 8 hours']

# Get counts
sleep_duration_counts = ( sleep_and_mood_df['Sleep_Duration'].value_counts().reindex(order, fill_value=0)
)

# Set style
sns.set_style("whitegrid")

plt.figure(figsize=(6,4))

# Create barplot
ax = sns.barplot( x=sleep_duration_counts.index, y=sleep_duration_counts.values, palette="viridis" )

# Add value labels on bars
for i, value in enumerate(sleep_duration_counts.values):
    ax.text(i, value + 0.5, str(value),
            ha='center', fontweight='bold')

plt.title('Distribution of Sleep Duration Among Respondents', fontsize=9, fontweight='bold')
plt.xlabel('Sleep Duration Category', fontsize=11)
plt.ylabel('Number of Respondents', fontsize=11)

plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



Analytical statement: The majority of respondents (49 participants) reported sleeping 6 to 8 hours, followed by 38 participants sleeping more than 8 hours. Only 14 respondents slept 4 to 6 hours, and very few (1 respondent) reported sleeping less than 4 hours. This indicates that most individuals in the dataset maintain a healthy sleep duration, with very limited cases of extreme sleep deprivation.

Q2. How does Sleep Duration affect Mood Rating? - Bivariate Analysis

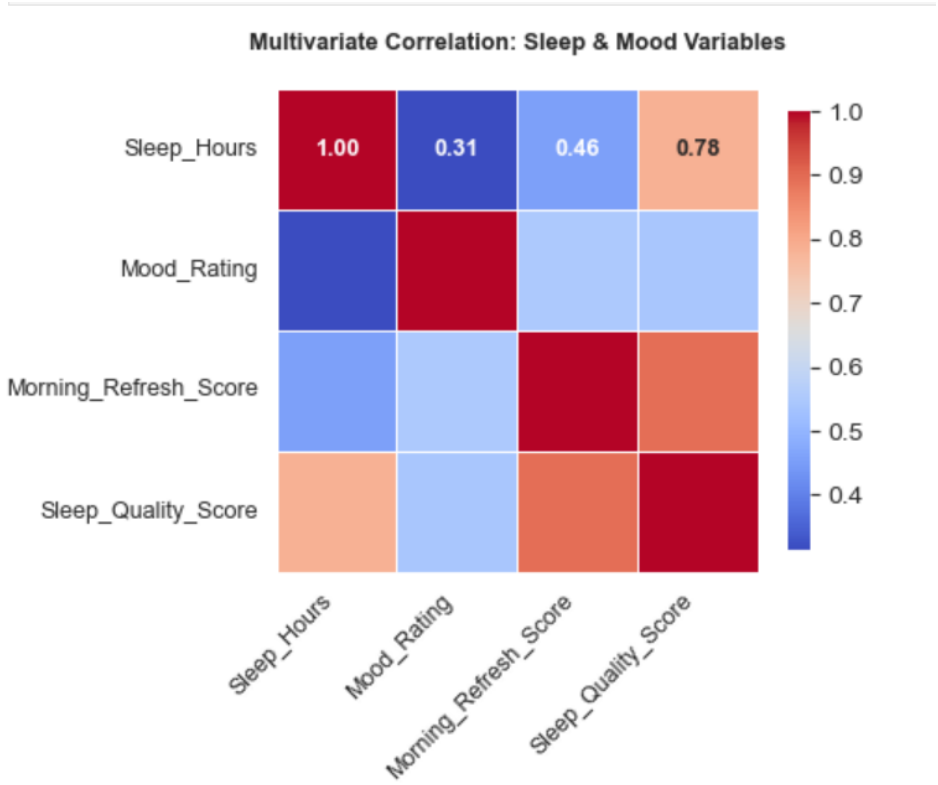
In [39]:
```
plt.figure(figsize=(6,4))

sns.barplot(
    x='Sleep_Duration',
    y='Mood_Rating',
    data=sleep_and_mood_df
)

plt.title('Average Mood Rating by Sleep Duration')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



Analytical statement: Mood rating increases with better sleep duration. Respondents who slept more than 8 hours reported the highest average mood rating (4.3), followed by those who slept 6 to 8 hours (4.0). Participants sleeping less than 4 hours showed the lowest mood rating (3.0). This suggests a positive relationship between sleep duration and mood, indicating that adequate sleep contributes to better emotional well-being.

Q3. What is the correlation between Sleep Hours, Mood Rating, Morning Refresh Score, and Sleep Quality Score? - Multivariate Analysis

In [40]:
```
# Select required columns
corr_data = sleep_and_mood_df[ ['Sleep_Hours','Mood_Rating','Morning_Refresh_Score','Sleep_Quality_Score'] ]

# Compute correlation
correlation_matrix = corr_data.corr()

# Set style
sns.set_style("white")

plt.figure(figsize=(5,5))

heatmap = sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap= "coolwarm",
    fmt=".2f",
    linewidths=0.5,
    square=True,
    cbar_kws={"shrink": 0.6},
    annot_kws={"size": 9, "weight": "bold"}
)

plt.title( "Multivariate Correlation: Sleep & Mood Variables", fontsize=9, fontweight='bold',pad=15 )

plt.xticks(rotation=45, ha='right', fontsize=9)
plt.yticks(fontsize=9)

plt.tight_layout()
plt.show()
```



Analytical statement: The heatmap reveals:

- Strong positive correlation between Sleep Hours and Sleep Quality Score (0.78)

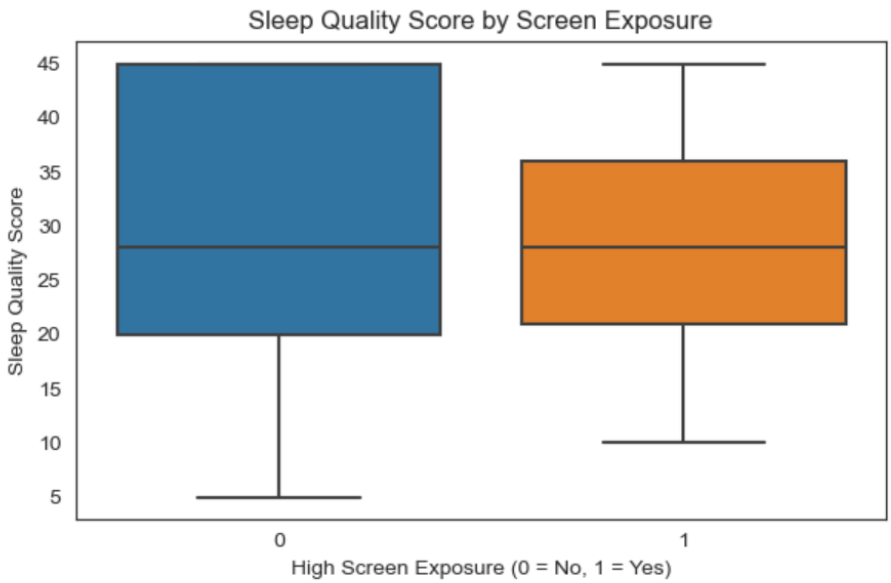
- Moderate positive correlation between Sleep Hours and Morning Refresh Score (0.46)
- Moderate positive correlation between Sleep Hours and Mood Rating (0.31)

Additionally, Morning Refresh Score and Sleep Quality Score show a strong positive relationship.

This indicates that longer sleep duration significantly improves sleep quality and refreshment levels, which in turn positively influences mood.

Q4. Does High Screen Exposure before bed reduce Sleep Quality Score? - Behavioural Pattern Analysis

In [41]:
```
plt.figure(figsize=(6,4))

sns.boxplot(
    x='High_Screen_Exposure',
    y='Sleep_Quality_Score',
    data=sleep_and_mood_df
)

plt.title('Sleep Quality Score by Screen Exposure')
plt.xlabel('High Screen Exposure (0 = No, 1 = Yes)')
plt.ylabel('Sleep Quality Score')

plt.tight_layout()
plt.show()
```

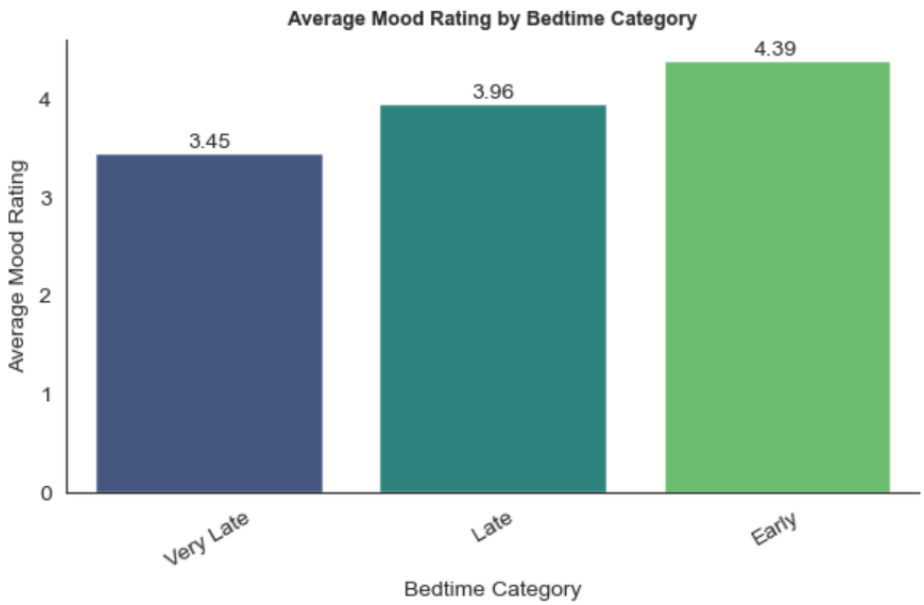


Analytical statement: The boxplot shows that individuals with high screen exposure before bedtime (1) tend to have slightly lower median sleep quality scores compared to those with low screen exposure (0).

- There is also greater variability in sleep quality among high screen users.
- This suggests that excessive screen exposure before sleep may negatively impact sleep quality.

Q5. Which Bedtime Category is associated with the lowest Mood Rating? - Insight-Driven Visualization

In [42]:
```
# Calculate mean mood rating per category and sort
bedtime_mood = ( sleep_and_mood_df.groupby('Bedtime_Category')['Mood_Rating'].mean().sort_values() )

plt.figure(figsize=(6,4))

# styling barplot
ax = sns.barplot( x=bedtime_mood.index, y=bedtime_mood.values, palette="viridis" )

# Add value labels on bars
for i, value in enumerate(bedtime_mood.values):
    plt.text(i, value + 0.05, f"{value:.2f}", ha='center')

plt.title('Average Mood Rating by Bedtime Category', fontsize=9, fontweight='bold')
plt.xlabel('Bedtime Category')
plt.ylabel('Average Mood Rating')
plt.xticks(rotation=30)

sns.despine()  # Clean look
plt.tight_layout()
plt.show()
```

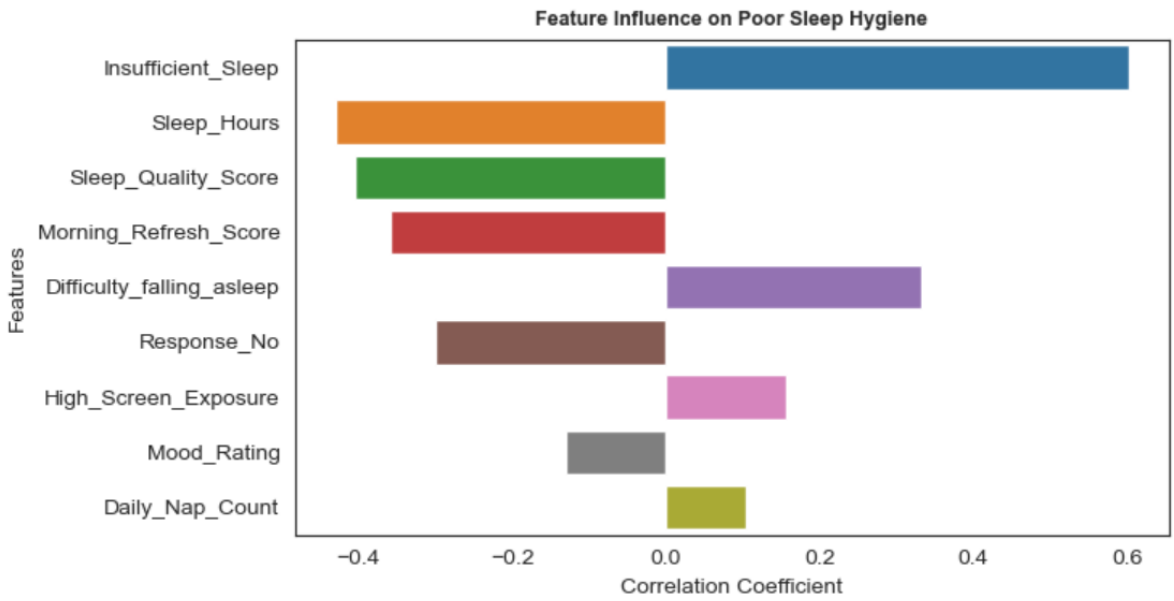


Analytical statement: Participants who went to bed early reported the highest average mood rating (4.39), followed by those who slept late (3.96).

- Those who went to bed very late had the lowest mood rating (~3.45).
- This indicates that earlier bedtime habits are associated with better mood outcomes, reinforcing the importance of consistent sleep schedules.

Q6. Which features most strongly influence Poor Sleep Hygiene? - Predictive Insight Visualization

In [43]:
```
# Correlation directly
corr = final_df.corr(numeric_only=True)['Poor_Sleep_Hygiene']

# Remove self-correlation
corr = corr.drop('Poor_Sleep_Hygiene')

# Sort by absolute strength
corr = corr.reindex(corr.abs().sort_values(ascending=False).index)

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(7,4))
sns.barplot(x=corr.values, y=corr.index)
plt.title('Feature Influence on Poor Sleep Hygiene', fontsize=9, fontweight='bold')
plt.xlabel('Correlation Coefficient')
plt.ylabel('Features')
plt.show()
```



Analytical statement: The strongest positive correlation with poor sleep hygiene is Insufficient Sleep (0.60), followed by Difficulty Falling Asleep (0.33) and High Screen Exposure (0.18).

- Negative correlations are observed with: Sleep Hours, Sleep Quality Score, Morning Refresh Score
- This suggests that poor sleep hygiene is primarily driven by insufficient sleep and sleep difficulties, while higher sleep quality and adequate sleep duration reduce the likelihood of poor sleep hygiene.

Conclusion

This project analysed the relationship between sleep duration, mood, sleep quality, screen exposure, and bedtime habits using exploratory data analysis and correlation techniques. Most respondents slept 6–8 hours, indicating generally healthy sleep patterns.

The results showed a clear positive relationship between sleep duration and mood. Longer sleep was strongly associated with better sleep quality, higher morning refresh scores, and improved mood. In contrast, insufficient sleep and difficulty falling asleep were the strongest contributors to poor sleep hygiene.

High screen exposure before bedtime was linked to slightly lower sleep quality, while early bedtime habits were associated with better mood outcomes.

Overall, the study highlights that adequate sleep, limited screen time before bed, and consistent sleep schedules are essential for maintaining good sleep hygiene and emotional well-being.

Overall, the study concludes that:

- Adequate sleep duration significantly improves sleep quality and mood.
- Early bedtime habits positively influence emotional well-being.
- High screen exposure before bedtime may negatively impact sleep quality.
- Insufficient sleep is the strongest predictor of poor sleep hygiene.

These findings reinforce the importance of maintaining healthy sleep practices to enhance overall well-being and mental health.